

Statistical modeling

Module 07

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Regression assumptions

Inference methods (ANOVA, F -tests, coefficient tests, confidence intervals) are all based on some theoretical results. They are valid under probabilistic assumptions about the model.

Departure from these assumptions may have consequences, in particular regarding all the inference:

- ▶ Prediction and confidence intervals could be incorrect,
- ▶ Tests and model comparisons could be not valid.

In order to know if our inferential conclusions are correct, we need to be able to check these assumptions.

Note: the estimation (OLS) is not so dependent on these assumptions. Somehow, a good model can be built with the OLS, even when these assumptions are not validated.

Checking assumptions of the regression

Consider a general model

$$y_i = b_0 + b_1x_{1i} + \cdots + b_px_{pi} + e_i$$

The assumptions about the residual errors e_i are:

- ▶ Mean 0,
- ▶ Variance σ^2 is the same for all e_i (in particular, in all groups, for any value of x , etc.),
- ▶ They are independent, in particular $cov(e_i, e_{i'}) = 0$,

This is overall summarized saying that the residual errors e_i are independent and identically distributed (i.i.d.) according to a normal distribution with mean 0 and variance σ^2 . It is written as

$$e_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$$

Regression assumptions

The residuals e_i are not directly observed. There are predicted after that the regression is fitted:

$$\hat{e}_i = y_i - \hat{y}_i$$

Under the model assumptions, $\hat{e}_i$ should be close to $N(0, \sigma^2)$, and in particular

- ▶ they should have a mean 0, independent of x .
- ▶ they should have a constant variance, independent of x .
- ▶ they should be normally distributed.

Standardized residuals

However, the $\hat{e}_i$ are not exactly $N(0, \sigma^2)$. Indeed, using the matrix notation

$$\hat{e} = y - \hat{y} = y - X\hat{b} = y - X(X^T X)^{-1} X y = (I - X(X^T X)^{-1} X) y = P y.$$

P has some properties like $P^2 = P^T P = P$. It can be shown that the so-called standardized residuals, defined as

$$r_i = \frac{\hat{e}_i}{s\sqrt{P_{ii}}},$$

are $N(0, 1)$, if the model assumptions are correct.

Therefore, for the diagnostics, r_i are often preferred to $\hat{e}_i$.

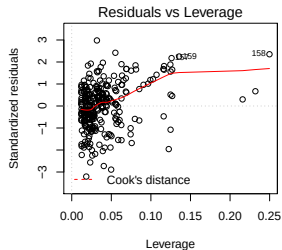
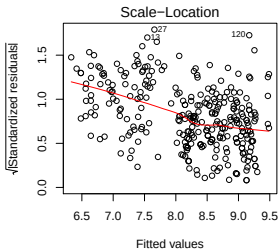
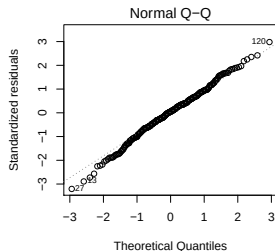
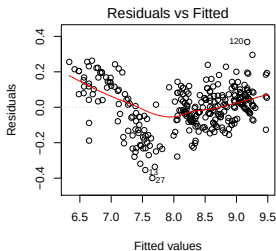
Regression assumptions: 4 graphs in R

To inspect the assumptions on the residuals, several methods exist. A quite popular one is based on specific graphs which helps to make a visual inspection of the residuals to check if the assumptions are satisfied.

It should be noted that these graphs give a general idea. It is not a 0–1 rule.

The default in R is to produce 4 plots.

Regression assumptions: 4 graphs in R



Regression assumptions: 4 plots in R

Top-left: Residuals $\hat{\epsilon}_i$ (y-axis) versus fitted values $\hat{y}_i$ (x-axis).

- ▶ Used to check the "0-mean" assumption:
- ▶ Red line is a visual help.
- ▶ Perfect match with the assumptions: the points should be scattered around the red line, and the red line should be flat.

Regression assumptions: 4 plots in R

Bottom-left: The square root of the absolute value of the standardized residuals $\sqrt{|r_i|}$ (y-axis) versus fitted values $\hat{y}_i$ (x-axis).

- ▶ Used to check the "constant variance" assumption σ^2 .
- ▶ Red line is a visual help.
- ▶ Perfect match with the assumptions: the points should be scattered around the red line, and the red line should be flat.

Regression assumptions: 4 plots in R

Top-right: QQ-plot of the standardized residuals r_i (y -axis) versus the normal distribution quantiles (x -axis).

If $r_i \sim N(0, 1)$, then the proportion of points among $\{r_1, \dots, r_n\}$ that are below z_α should be close to α , for any $0 < \alpha < 1$. This translates into a graph of

- ▶ on the x -axis, $z_{1/(n+1)}, \dots, z_{n/(n+1)}$,
- ▶ on the y -axis, the ordered values $r_{(1)} \leq \dots \leq r_{(n)}$.

These two should match around the $1 - 1$ diagonal.

Regression assumptions: 4 plots in R

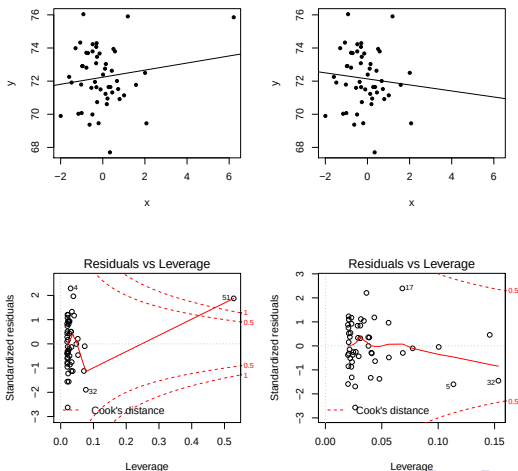
Bottom-right: The standardized residuals r_j versus a leverage index of the corresponding observation (x -axis).

- ▶ Use to detect an influential observation.
- ▶ Influential bounds are in dotted lines (outside the graph range here).
- ▶ An influential data point appears outside these bounds. Such points should be inspected and maybe removed from the data base.

The leverage index explanation is outside the scope of the course.

Regression assumptions: 4 plots in R

The right-most point without being an outlier is influencing a lot the regression line estimation. Its influence is detected by the diagnostic graph. Removing it is a case-by-case issue.



Checking assumptions of the regression: model aptness

Residual analysis is also called model "aptness" analysis.

- ▶ These diagnostics, though technical, are only visual help.
- ▶ They should not be over-interpreted: no black-or-white conclusion.
- ▶ They require a statistical computer program (technical).
- ▶ Often, even when the diagnostics are not completely satisfactory, the model analysis (estimation and inference) remains valid to some extent.
- ▶ These diagnostics are used for general conclusions and also for detecting problematic data points (graph 4).

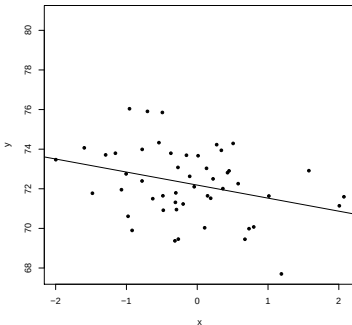
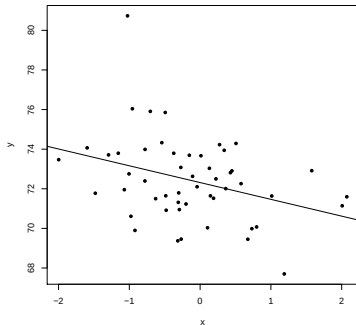
Checking assumptions of the regression: model aptness

In practice,

- ▶ The model is fitted to the data.
- ▶ The analysis of the residuals is performed on the model.
- ▶ If the diagnostics are OK, then the analysis is pursued (tests, ANOVA, etc.).
- ▶ If the diagnostics are very bad, then actions should be taken:
 - ▶ Make Changes of variables (e.g. $\log y$, $\log x$, etc.), or by adding factors to the model (e.g. x^2 , etc.). It is a case-by-case issue.
 - ▶ Some non-parametric alternative exist.
- ▶ If the diagnostics are "average", then the analysis is pursued with some extra caution in particular to p -values that are close to 5%.

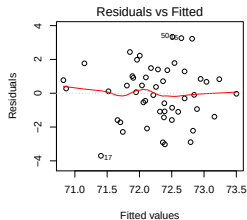
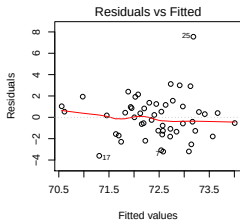
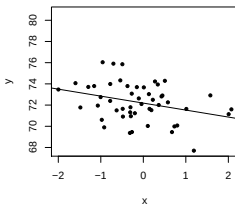
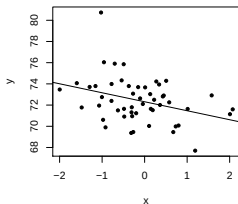
Outliers

The definition of an outlier is quite subjective. It is an observation which, in some ways, is far from what we can expect. Below, observation 25 could be considered as unexpected.



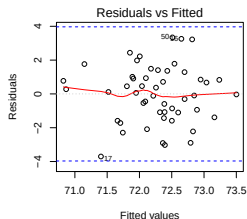
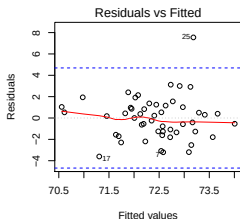
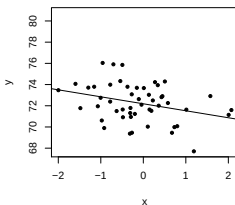
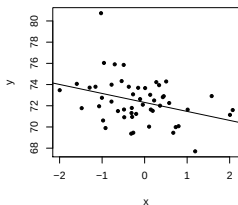
Outliers

One way to detect outliers is to fit a regression line and to inspect the residuals (diagnostic graph 1).



Outliers

In this context, one formal definition of an outlier (with respect to the regression) is an observation such that its residual is outside a range of 99% of the normal distribution.



Outliers

Note that, with this definition, being an outlier is not a "negative" characteristic. It just means that it was unlikely¹ to be observed.

The choice of the level 99% was convenient for this example, because there are 51 observations. However, if there are $n = 200$ observations, then we expect to see about 2 outliers.

In this course, we set the level of outliers detection to about $1/(2n)$. Thus, we expect to see less than 1 outlier.

In any case, action to take after an outlier detection is a subjective decision.

¹unlikely according to the model.

Outliers: discussion

- ▶ Regression is a powerful mean to analyze outliers because it can be performed even in large dimension where scatter-plots cannot helps.
- ▶ Influential observation and outliers are not synonymous. An outlier can be very influential. An influential observation need not be an outlier.
- ▶ The presence of influential outliers is problematic since the model may be so badly fitted that the outlier cannot be detected from the residuals.
- ▶ Removing an outlier should not be systematic. In general, its source should be analyzed (error of observation, data contamination, etc.).
- ▶ Robust statistics is the science of developing statistical methods which are robust to the presence of influential observations and outliers.